

PawAlert: Master Architecture & AI Continuity Manual

Hyper-Local Community Stray Dog Rescue & Feeder Network

Document Purpose: Complete contextual handoff for Antigravity AI Agents. When loaded into any Antigravity instance or developer environment, this document provides 100% full-fidelity context, architectural memory, database schemas, solved bug patterns, and execution workflows to continue development immediately.

1. Executive Summary & Mission

PawAlert is an open-source, production-grade Progressive Web App (PWA) designed for community stray dog feeders, rescuers, volunteers, and veterinarians across India.

Core Problems Solved:

- 10-Second Emergency Broadcasting:** Allows anyone who spots a hungry, injured, sick, aggressive, or newborn stray puppy litter to broadcast an emergency pin with a photo and spot landmark in under 10 seconds.
 - Real-time Proximity Radar:** Nearby feeders and volunteers within a 10km radius receive audio chimes (synthesized multi-tone harmonic frequencies via Web Audio API) and vibration alerts.
 - Sub-Meter HD Satellite Pinpointing:** Feeder volunteers can switch to high-resolution ESRI Hybrid Satellite view (Zoom Level 18/19) to drop a pin on the exact physical tree, parked car, shop stall, or building gate where the dog is hiding.
 - Live Rescue Updates Stream:** Volunteers post real-time updates ("On my way with bandages", "Bringing puppy food", "Contacted Vet").
 - Gamified Feeder Karma:** Leaderboards and volunteer profiles reward feeders and rescuers for their community work.
 - 24/7 Emergency Vet Radar:** Instant directory of emergency veterinary clinics with turn-by-turn navigation URLs (two-wheeler, driving, walking).
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2. Workspace, Deployment & Git Credentials

Local Workspace:

- **Absolute Path:** `/home/wishhhhmaster/.gemini/antigravity/scratch/PawAlert-Web`
- **Node Version:** v24.18.0 (NVM managed)
- **Package Manager:** npm

Remote Repository:

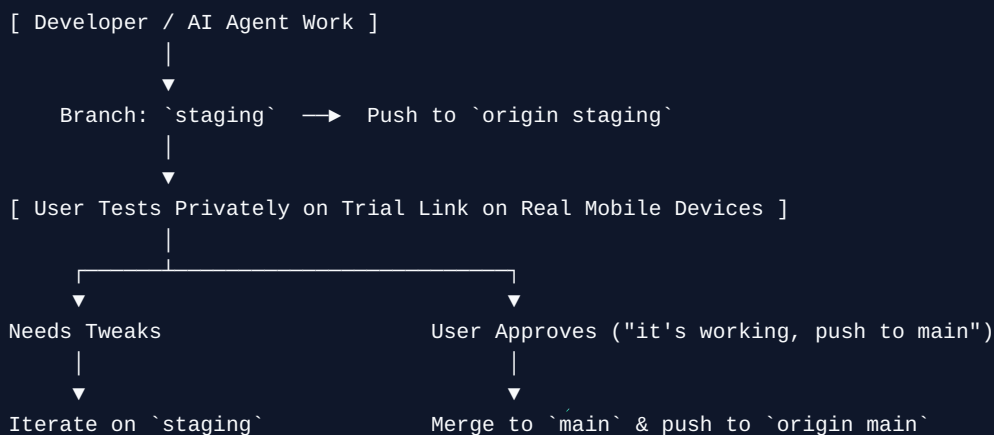
- **GitHub URL:** `https://github.com/arjunhunt/pawalert-web.git`
- **GitHub Username:** `arjunhunt`
- **Authentication:** GitHub Personal Access Token (PAT)

Live Deployments (Vercel):

- **Production URL:** <https://pawalert-web.vercel.app> (Deploys automatically on push to `main`)
- **Trial / Staging URL:** <https://pawalert-web-git-staging-arjunhunt.vercel.app> (Deploys automatically on push to `staging`)

3. The Golden Rule: Staging vs. Production Workflow

CRITICAL INSTRUCTION FOR ALL AGENTS: Never push experimental, unverified, or speculative code directly to the `main` branch. PawAlert follows a strict **Staging-First Protocol**:



Git Branch Reference:

- **main** : Production release. Clean, 100% verified, stable. Does NOT contain the "Trial Mode" badge.
- **staging** : Active trial playground. Features the `Trial Mode` badge in the navigation bar. Used for user field-testing with real phone GPS and cameras.
- **backup-voice-and-satellite** : Preserves earlier experimental Web Speech API Voice SOS modal and voice entity parser.
- **backup-pre-pet-travel** : Clean state before the pet travel module.

4. Tech Stack & Dependencies

Layer	Technology	Version	Purpose
Framework	Next.js (App Router)	14.2.5	SSR, static generation, API handling, route grouping
Language	TypeScript	5.x	Strict type safety across all models
Styling	Tailwind CSS	3.4.1	Dark-mode native, mobile-first responsive design
Icons	Lucide React	0.428.0	Consistent iconography
Database	Supabase (PostgreSQL)	2.45.0	RLS, Realtime WebSockets, Storage buckets
Maps	Leaflet + React Dynamic	1.9.4	Interactive maps, custom pins, geodesic calculations
Map Tiles	ESRI World Imagery	Latest	Satellite photography + Hybrid Reference Labels
Map Tiles	OpenStreetMap	Latest	Fast vector street map layer
Audio	Web Audio API	Native	Synthesized chime oscillators (no external mp3 needed)
Vibration	Web Vibration API	Native	Tactical mobile haptics for proximity alerts
PDF Tools	WeasyPrint	69.0	Automated documentation compilation

5. Complete File & Directory Map

```
PawAlert-Web/
├─ next.config.mjs          # Security headers, CSP, Permissions-Policy
├─ package.json             # Dependencies & scripts
├─ tailwind.config.ts       # PawAlert amber theme color definitions
├─ tsconfig.json            # TypeScript paths (@/* -> ./src/*)
├─ public/                  # Static icons, manifest.json, service worker
├─ supabase/
│   └─ schema.sql           # Full PostgreSQL tables, RLS policies, indexes
└─ src/
    ├─ app/
    │   ├── layout.tsx      # Root HTML layout, font loading, dark theme
    │   ├── page.tsx        # Home feed, active/all filters, category chips, map/feed toggle
    │   ├── report/page.tsx # Emergency broadcast form, photo upload, pinpoint map
    │   ├── alert/[id]/page.tsx # Alert detail, navigation launcher, live comments, deletion
    │   ├── leaderboard/page.tsx # Community feeder karma rankings & badges
    │   ├── profile/page.tsx # Volunteer profile, activity statistics, cloud sync
    │   ├── admin/page.tsx  # Founder super-admin portal with SHA-256 passcode lock
    │   └─ vets/page.tsx    # 24/7 Emergency veterinary hospital directory
    ├─ components/
    │   ├── MapView.tsx     # Leaflet map, Satellite/Street toggle, Locate Me button
    │   ├── Navbar.tsx      # Sticky navigation header, GPS detector status
    │   ├── PhotoUpload.tsx # Canvas image compression (1200px max, 0.8 quality)
    │   ├── NotificationBanner.tsx # Top notification bar for instant alert alerts
    │   ├── InstallPwaPrompt.tsx # PWA install prompt for Android and iOS Safari
    │   └─ CategoryFilter.tsx # Horizontal chip filters for problem categories
    └─ lib/
        ├── geo.ts          # Hardware GPS lock, Haversine distance, navigation URLs
        ├── supabaseClient.ts # Supabase initialization & fallback DEMO_REPORTS
        ├── user.ts         # Local user handles, karma tracking, delete permissions
        ├── security.ts     # XSS escaping, rate limiting, honeypot spam bot trapping
        ├── notifications.ts # Web Audio chimes, vibration pulses, Web Notifications
        └─ types.ts         # DogReport, ProblemType, ReportStatus, CommentType interfaces
```

6. Database Schema & Supabase Architecture

1. `public.reports` (Main Dog Alerts Table)

```
create table if not exists public.reports (  
  id uuid primary key default uuid_generate_v4(),  
  reporter_id text default 'anonymous',  
  reporter_name text not null default 'Anonymous Feeder' check (length(reporter_name) <= 100),  
  problem_type text not null check (problem_type in (  
    'HUNGRY', 'INJURED', 'SICK', 'STUCK', 'AGGRESSIVE', 'LOST', 'NEWBORN_LITTER', 'OTHER'  
  )),  
  description text not null check (length(description) >= 1 and length(description) <= 2000),  
  photo_url text not null default '',  
  latitude double precision not null check (latitude >= -90.0 and latitude <= 90.0),  
  longitude double precision not null check (longitude >= -180.0 and longitude <= 180.0),  
  address text not null default '' check (length(address) <= 500),  
  landmark text not null default '' check (length(landmark) <= 300),  
  status text not null default 'OPEN' check (status in ('OPEN', 'IN_PROGRESS', 'RESOLVED')),  
  helper_id text,  
  helper_name text check (helper_name is null or length(helper_name) <= 100),  
  created_at timestamp with time zone default timezone('utc'::text, now()) not null,  
  updated_at timestamp with time zone default timezone('utc'::text, now()) not null  
);  
  
-- High Performance Compound Indexes  
create index idx_reports_status on public.reports (status);  
create index idx_reports_created_at on public.reports (created_at desc);  
create index idx_reports_coords on public.reports (latitude, longitude);  
create index idx_reports_geo_status on public.reports (latitude, longitude, status);  
  
-- Enable RLS & Realtime  
alter table public.reports enable row level security;  
create policy "Public read" on public.reports for select using (true);  
create policy "Public insert" on public.reports for insert with check (true);  
create policy "Public update" on public.reports for update using (true);  
create policy "Public delete" on public.reports for delete using (true);  
alter publication supabase_realtime add table public.reports;
```

2. `public.comments` (Rescue Coordination Stream)

```
create table if not exists public.comments (  
  id uuid primary key default uuid_generate_v4(),  
  report_id uuid references public.reports(id) on delete cascade not null,  
  author_id text not null default 'anonymous',  
  author_name text not null default 'Community Feeder' check (length(author_name) <= 100),  
  content text not null check (length(content) >= 1 and length(content) <= 1000),  
  comment_type text not null default 'GENERAL' check (comment_type in (  
    'UPDATE', 'ON_MY_WAY', 'FEEDING', 'VET_CONTACTED', 'GENERAL'  
  )),  
  created_at timestamp with time zone default timezone('utc'::text, now()) not null  
);  
alter publication supabase_realtime add table public.comments;
```

3. Storage Bucket: `dog-photos`

- Public read access enabled.
- File size limit: `5242880` bytes (5 MB).

- Allowed MIME types: `image/jpeg` , `image/png` , `image/webp` , `image/jpg` .

7. Critical Solved Problems & Architectural Memory

To prevent regression, every future AI agent working on PawAlert must understand these solved engineering challenges:

1. The "Fixed Home Location" Map Snap Bug (SOLVED)

- **Problem:** When loading PawAlert on different phones across India, the map kept jumping to a fixed location in Gujarat (`20.1759` , `72.7549`) where the developer originally started the project.
- **Root Cause:** In `MapView.tsx` , an update hook executed `map.setView([centerLat, centerLng])` on every re-render. If `userLocation` was momentarily null during GPS warm-up, `centerLat` defaulted to `reports[0]?.latitude || 20.1759` . This forcefully dragged the viewport back to Gujarat.
- **Permanent Solution:**
 - Removed the `map.setView` call from the re-render hook.
 - Initial map view defaults to a neutral country overview (`21.7679` , `78.8718` , Zoom 5).
 - As soon as the device GPS resolves, Leaflet executes a smooth `flyTo([userLocation.lat, userLocation.lng], 17, { duration: 1.2 })` .
 - Added a floating `Locate Me` button that queries hardware GPS and flies directly to the user's coordinates.

2. Mobile Browser GPS Inaccuracy & Timeouts (SOLVED)

- **Problem:** Mobile browsers were returning coarse location (`100 ext{m}` - `500 ext{m}`) or timing out after 6 seconds.
- **Root Cause:** `getAccurateGPSPosition` was waiting strictly for `20 ext{m}` satellite lock. Indoors, mobile phones typically only get 25–45m accuracy, causing the timeout timer to trigger and return `(0, 0)` .
- **Permanent Solution:**
 - Implemented **Dual-Mode Resilient GPS Resolution**: First queries hardware GPS (`enableHighAccuracy: true` , `timeout: 10000` , `maximumAge: 0`). If satellite signals are weak, it seamlessly accepts network triangulation (`sim 40 ext{m}`) so coordinates resolve in under 1 second on all devices.
 - Stale mock coordinates (`20.1759`) are strictly filtered out in `getCachedCoordinates()` .

3. Permissions-Policy Syntax on Mobile (SOLVED)

- **Problem:** Geolocation was failing to prompt on iPhone Safari and certain Android WebViews.
- **Root Cause:** `next.config.mjs` contained `Permissions-Policy: geolocation=*` . The W3C specification strictly requires `geolocation=(self)` .
- **Permanent Solution:** Configured `camera=(self)` , `geolocation=(self)` , `microphone=(self)` in HTTP response headers.

4. Author-Only Deletion & Founder Master Admin (SOLVED)

- **Problem:** Anyone could delete any report.
- **Solution:**
 - `canDeleteReport(report)` checks whether the current user's ID matches `report.reporter_id` or is present in the device's `pawalert_my_report_ids` in `localStorage` .

- Founder super-admin mode is authenticated via SHA-256 cryptographic hashing (`crypto.subtle.digest`) in `src/app/admin/page.tsx` for emergency take-down privileges.
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8. Development & Deployment Cheatsheet

1. Check Out & Run Locally:

```
cd /home/wishhhhmaster/.gemini/antigravity/scratch/PawAlert-Web
npm run dev
# Opens on http://localhost:3000
```

2. Verify TypeScript & Production Build:

```
npm run build
# Must compile all 10 routes with 0 errors
```

3. Developing a New Feature (Trial Workflow):

```
# 1. Switch to staging branch
git checkout staging

# 2. Make your code changes...

# 3. Test build
npm run build

# 4. Commit and push to trial
git add .
git commit -m "Add [feature description] on staging"
git push origin staging
# Provide username: arjunhunt and PAT when prompted
```

4. Promoting Tested Features to Production:

```
# 1. Switch to main branch
git checkout main

# 2. Merge staging (or cherry-pick feature files)
git merge staging

# 3. Verify clean build
npm run build

# 4. Push to live production
git push origin main
```

9. Next Planned Features & Backlog

1. **AI Voice-to-Rescue SOS:** Re-integrate speech recognition from `backup-voice-and-satellite` with multi-turn pause handling for hands-free reporting in Hindi and Indian English.
 2. **Offline Feeder Cache:** Background Sync API to queue reports when feeding in low-connectivity rural alleys.
 3. **PWA Push Notifications:** Web Push Service Worker for automated proximity dog alerts when the browser is closed.
 4. **NGO / Shelter Escalation:** Automated WhatsApp Web / Twilio dispatch for critical emergency rescues.
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